

INFINITE CHOCOLATE BAR
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FOUR-PIECE SQUARE
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INTERMISSION
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$64 = 65$
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CONNECTION
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COOL FIGURES
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Investigating Geometric Paradoxes

September 5th, 2026

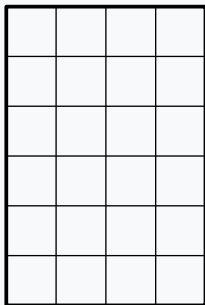
Jacob Khohayting

The Infinite Chocolate Bar

We have all (*maybe*) seen the Infinite Chocolate Bar Problem! ([Video](#))

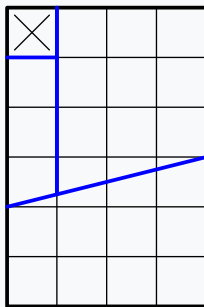
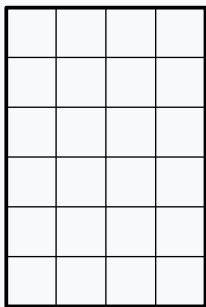
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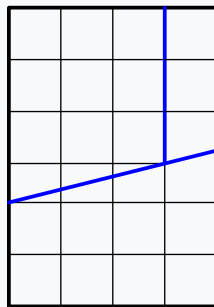
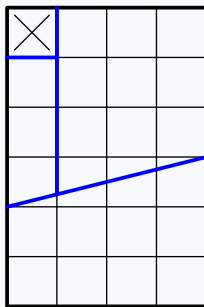
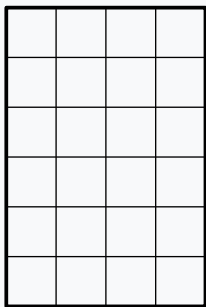
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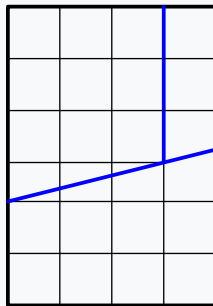
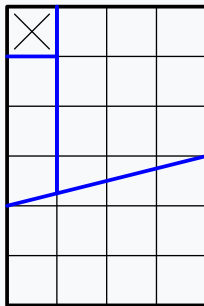
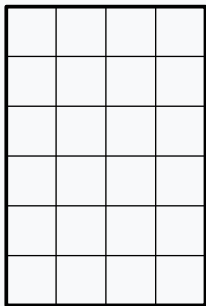
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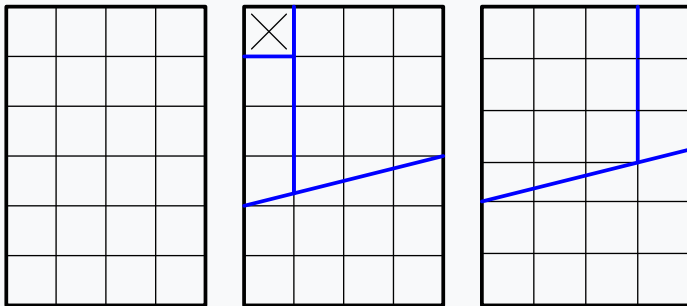
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Q:

The Infinite Chocolate Bar

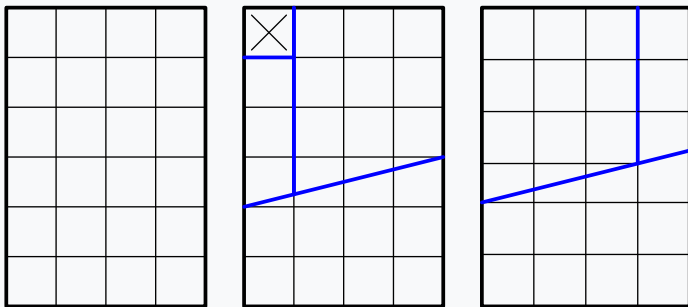
We have all (*maybe*) seen the Infinite Chocolate Bar Problem! ([Video](#))



Q: How does this work?

The Infinite Chocolate Bar

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Q: How does this work? Do we really get infinitely many pieces?

How does this work?

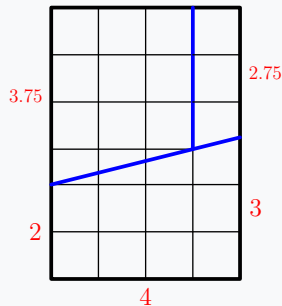
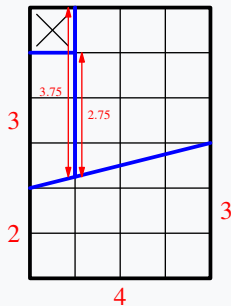
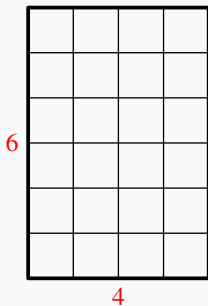
Answer:

How does this work?

Answer: No.

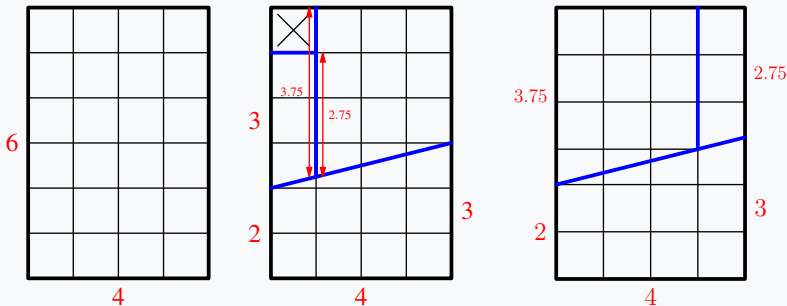
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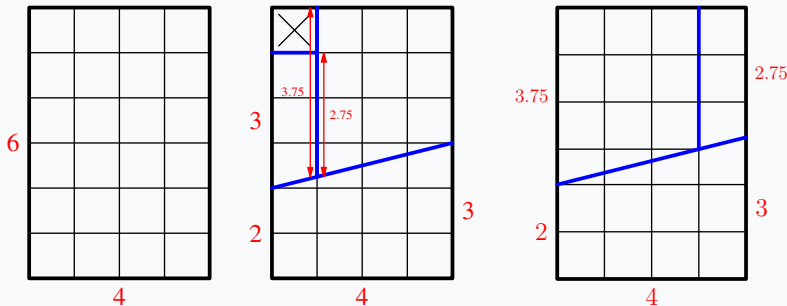
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Indeed, the 4×6 rectangular grid becomes a 4×5.75 rectangular grid.

How does this work?

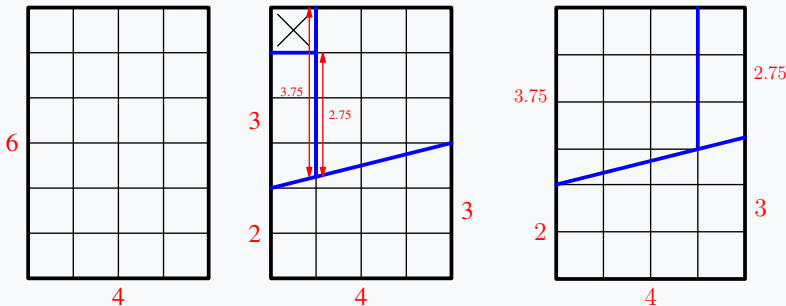
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Indeed, the 4×6 rectangular grid becomes a 4×5.75 rectangular grid. Therefore, initial grid has area 24 while the new grid has area 23.

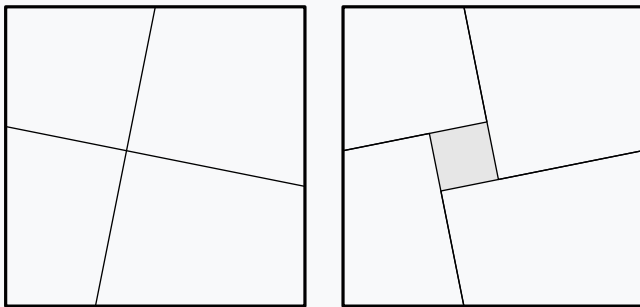
How does this work?

Answer: No.



Indeed, the 4×6 rectangular grid becomes a 4×5.75 rectangular grid. Therefore, initial grid has area 24 while the new grid has area 23. This makes sense as we remove a single square, and so a square of area 1.

Four-piece Square Illusion



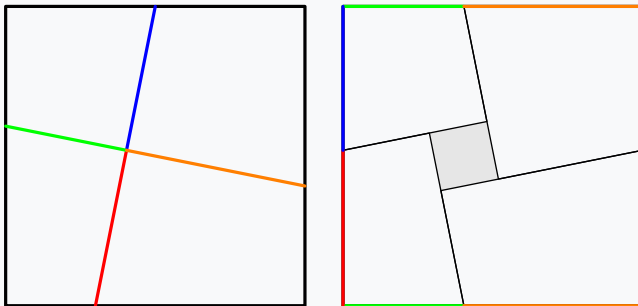
Two perpendicular lines dissect the square on the left to form four regions. Rotate each region as shown on the left figure to form a new square as shown on the right figure. A new smaller square is formed! How did this happen?

Four-piece Square Illusion

Under the rotation, the following sides map as follows:

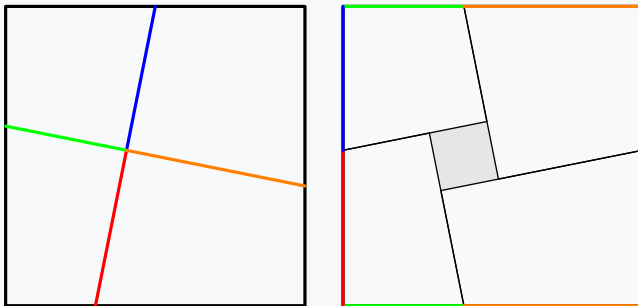
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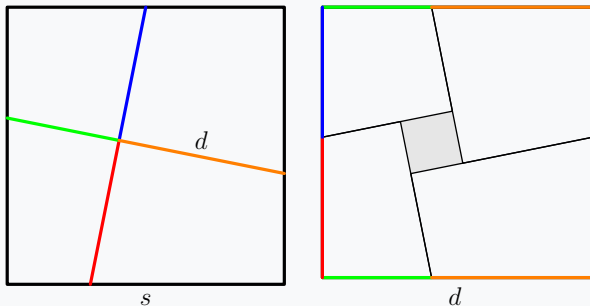
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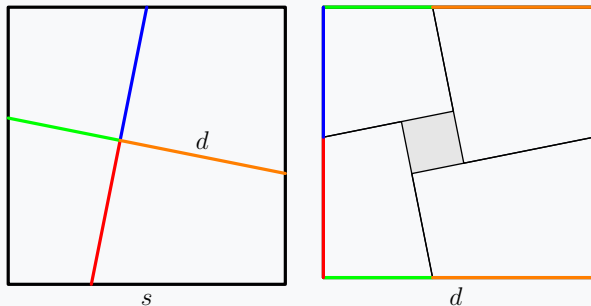
As a result, the side of the squares transform from the initial side to the diagonal side inside the square!

Four-piece Square Illusion



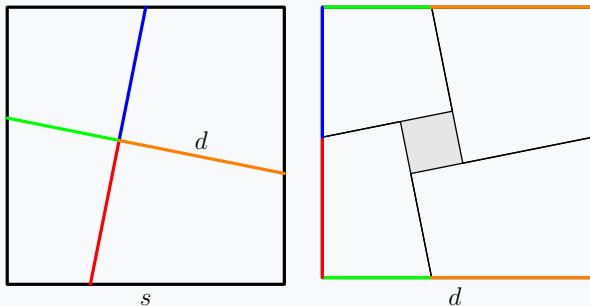
The area changes from s^2 to d^2 .

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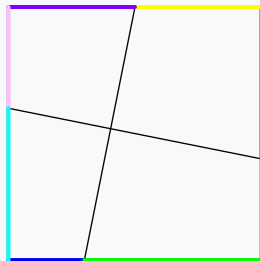
The area changes from s^2 to d^2 . Thus, the small square in the center must be $d^2 - s^2$.

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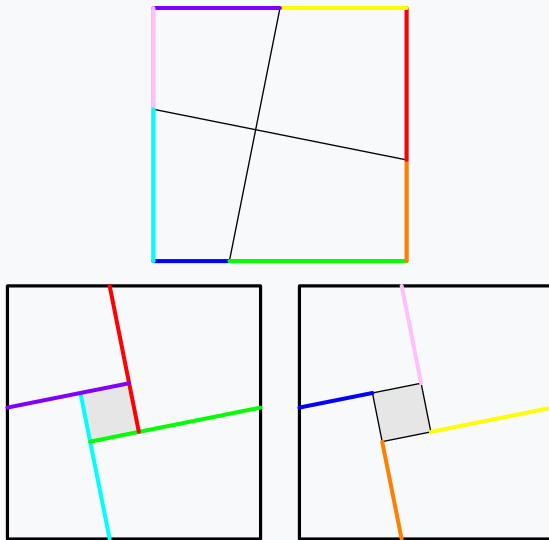


The area changes from s^2 to d^2 . Thus, the small square in the center must be $d^2 - s^2$. Let's dive into why this is true by consider how the side changes under the rotation.

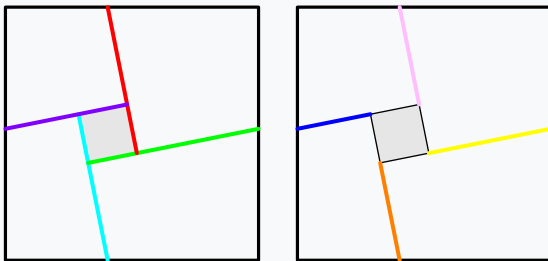
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Four-piece Square Illusion



It follows that the four sides are:

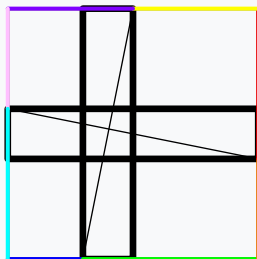
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- purple — blue
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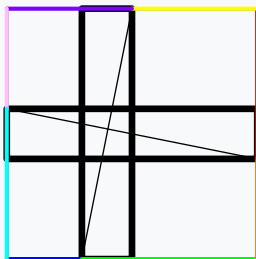
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All of these lengths are $\sqrt{d^2 - s^2}$ by Pythagorean Theorem. So, the inner square has area $d^2 - s^2$ which makes sense.

Fibonacci Numbers

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- $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$.

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Indeed, we can enumerate and find a bunch of terms in the sequence.

F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}
0	1	1	2	3	5	8	13	21	34	55	89	144

Properties of the Fibonacci Numbers

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Some of my favorites include

- $\sum_{k=1}^n F_k = F_{n+2} - 1.$
- $\sum_{k=1}^n F_{2k} = F_{2n+1} - 1.$
- $\sum_{k=1}^n F_{2k-1} = F_{2n}.$

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Let $\{F_n\}_{n \geq 0}$ be the sequence of Fibonacci Numbers. Then,

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Why is this true? Let's prove it.

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Base Case: $n = 1$

Note that $F_0 = 0$, $F_1 = 1$, and $F_2 = 1$. Therefore,

$$F_2F_0 - F_1^2 = -1 = (-1)^1,$$

and so the claim holds.

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Now, suppose we prove the claim for n . That is, suppose

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n$$

holds. We will show

$$F_{n+2}F_n - F_{n+1}^2 = (-1)^{n+1}$$

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Notice,

$$\begin{aligned} F_{n+2}F_n - F_{n+1}^2 &= (F_{n+1} + F_n)F_n - (F_n + F_{n-1})F_{n+1} \\ &= (F_{n+1}F_n + F_n^2) - (F_nF_{n+1} + F_{n-1}F_{n+1}) \\ &= F_n^2 - F_{n-1}F_{n+1} \\ &= -(-1)^n \\ &= (-1)^{n+1}, \end{aligned}$$

as desired.

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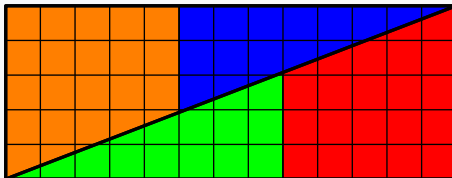
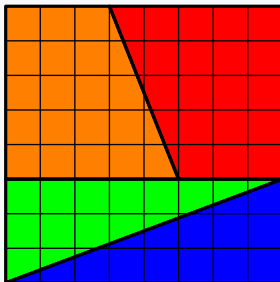
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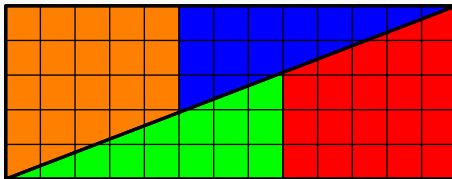
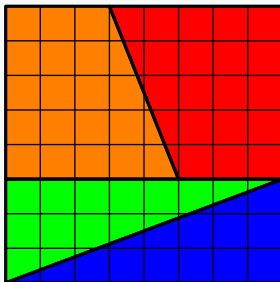
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as desired. Due to Induction, the claim holds for all n .

How does this dissection work?

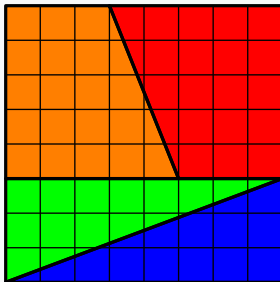


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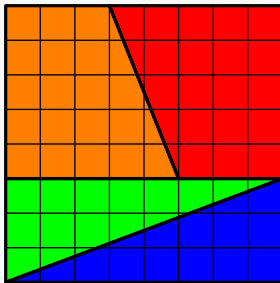


Q: How does this work? The figure grows by 1 square!

How does this dissection work?

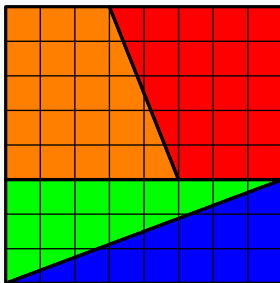


How does this dissection work?



Consider the slope of each slanted portion of each tile:

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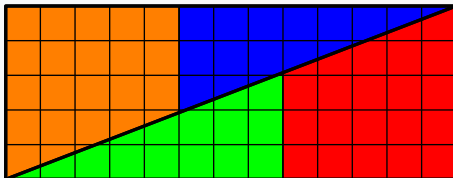
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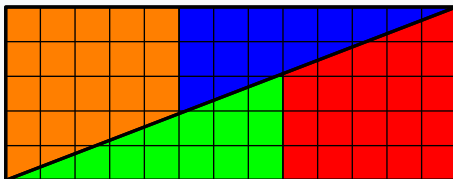
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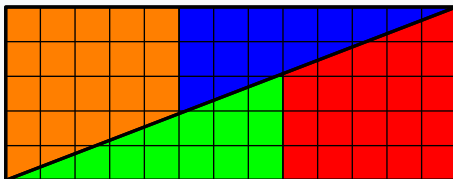


We expect these slopes to be $\frac{5}{13}$! Not $\frac{2}{5}$ or $\frac{3}{8}$!

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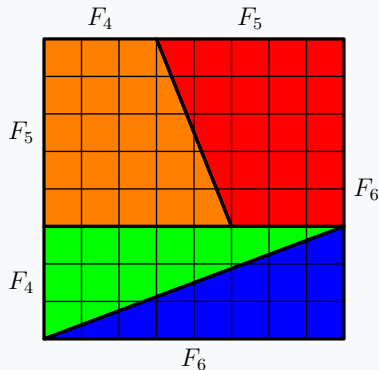


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How does this relate to our Fibonacci Sequence?

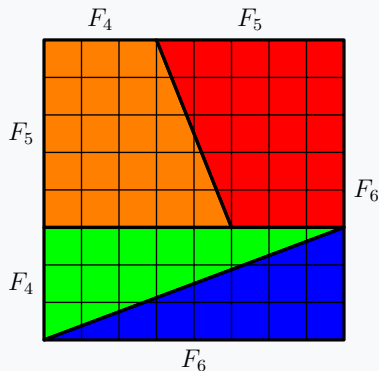
Relation to Fibonacci Sequence

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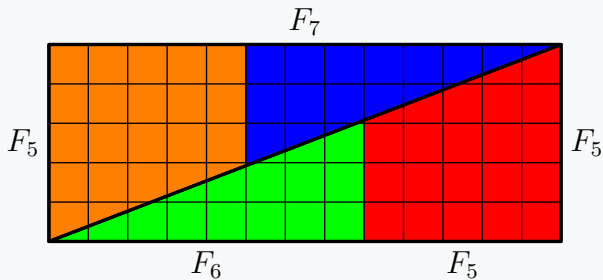
Consider the following:



The area of this figure is F_6^2 .

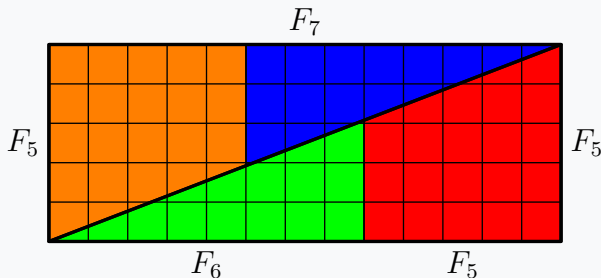
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The area of this figure is $F_5 F_7$.

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Recall:

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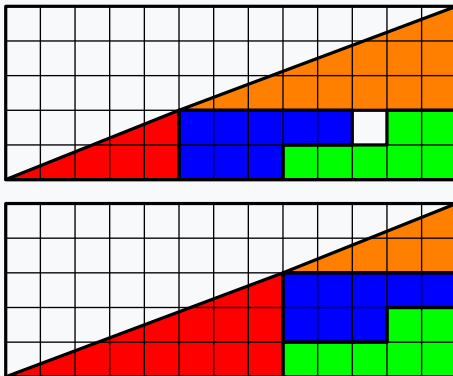
$$F_{n-1}F_{n+1} - F_n^2 = (-1)^n$$

It follows that

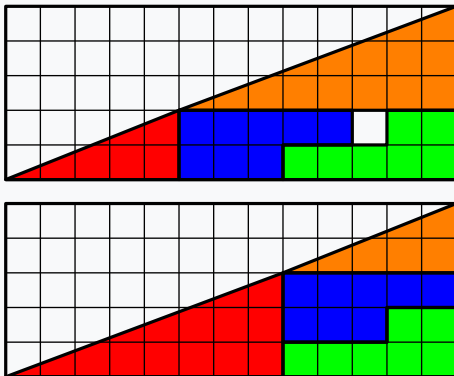
$$F_5F_7 - F_6^2 = (-1)^6 = 1.$$

So, the area does indeed increase by 1.

Curry Triangles

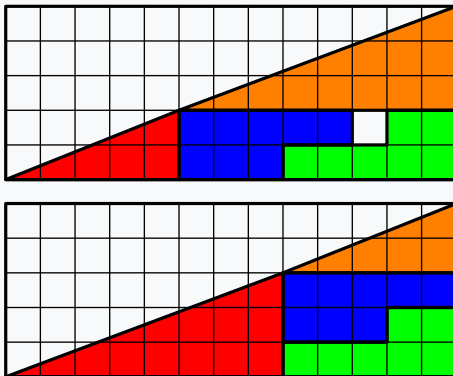


Curry Triangles



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Curry Triangles



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Recap



- When we select slopes of $\frac{F_{n-2}}{F_n}$ and $\frac{F_{n-1}}{F_{n+1}}$, the area changes by 1.
- $\left| \frac{F_{n-2}}{F_n} - \frac{F_{n-1}}{F_{n+1}} \right|$ is a unit fraction. In particular,

$$F_{n+1}F_{n-2} - F_nF_{n-1} = \pm 1.$$

- Can we make more versions? How about ones that don't have a slope of $\frac{F_{n-2}}{F_n}$?

Continued Fractions

We will look at the expression:

$$\begin{array}{l} 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots}}}}} \end{array}$$

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We usually abbreviate this by $[1; 1, 1, 1, 1, 1, 1, \dots]$.

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Does it look familiar?

Continued Fractions

In particular, we will cut it off and look at the “convergents”. That is,

$$\begin{aligned}[1] &= \frac{1}{1} \\[1; 1] &= 1 + \frac{1}{1} = \frac{2}{1} \\[1; 1, 1] &= 1 + \frac{1}{1 + \frac{1}{1}} = \frac{3}{2} \\[1; 1, 1, 1] &= 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}} = \frac{5}{3}\end{aligned}$$

Does it look familiar? **It looks like the Fibonacci Sequence!**

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 \end{aligned}$$

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Continued Fractions with F_n

In particular, we will cut it off and look at the “convergents”. That is,

$$[1] = \frac{1}{1} = \frac{F_2}{F_1}$$

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$$[1; 1, 1] = 1 + \frac{1}{1 + \frac{1}{1}} = \frac{3}{2} = \frac{F_4}{F_3}$$

$$[1; 1, 1, 1] = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}} = \frac{5}{3} = \frac{F_5}{F_4}$$

Does it look familiar? **It looks like the Fibonacci Sequence!** Why?

Continued Fractions with F_n

If

$$\underbrace{[1; 1, 1, 1, \dots]}_{n \text{ occurrences}} = \frac{F_{n+1}}{F_n},$$

then

$$\underbrace{[1; 1, 1, 1, \dots]}_{n+1 \text{ occurrences}} = 1 + \frac{1}{\underbrace{[1; 1, 1, 1, \dots]}_{n \text{ occurrences}}} = 1 + \frac{1}{F_{n+1}/F_n} = \frac{F_{n+2}}{F_{n+1}}.$$

Continued Fractions with F_n

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This is nice because these values are the slopes of the triangles. Can we generalize though?

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This is nice because these values are the slopes of the triangles. Can we generalize though? Is it true for any continued fraction that the difference between consecutive convergents yields a unit fraction? Will the difference in area always be 1 when we select two consecutive convergents as slopes?

Can we generalize?

Consider this:

$$\begin{aligned} [a_0; a_1, a_2, a_3, \dots] = & a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \frac{1}{a_5 + \frac{1}{a_6 + \dots}}}}} \end{aligned}$$

Can we generalize?

Define

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, a_3, \dots, a_n].$$

These are the “*convergents*”!

Recursive Formula

Theorem

If $\frac{p_n}{q_n}$ is the n th convergent to a continued fraction $[a_0, a_1, a_2, \dots]$. Then,

$$p_n = a_n \cdot p_{n-1} + p_{n-2}, \text{ and}$$

$$q_n = a_n \cdot q_{n-1} + q_{n-2}.$$

Here, $p_0 = a_0$, $p_1 = a_0 a_1 + 1$, $q_0 = 1$, and $q_1 = a_1$.

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We will prove this with induction. For $n = 0$ and $n = 1$:

$$\frac{p_0}{q_0} = \frac{a_0}{1}, \text{ and}$$

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$$\frac{p_1}{q_1} = a_0 + \frac{1}{a_1} = \frac{a_0 a_1 + 1}{a_1}.$$

Recursive Formula

For $n > 1$,

$$\frac{p_n}{q_n} = [a_0, a_1, \dots, a_n]$$

Recursive Formula

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Recursive Formula

For $n > 1$,

$$\begin{aligned}\frac{p_n}{q_n} &= [a_0, a_1, \dots, a_n] \\ &= \left[a_0, a_1, \dots, a_{n-1} + \frac{1}{a_n} \right] \\ &= \frac{\left(a_{n-1} + \frac{1}{a_n} \right) p_{n-2} + p_{n-3}}{\left(a_{n-1} + \frac{1}{a_n} \right) q_{n-2} + q_{n-3}}\end{aligned}$$

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Recursive Formula

For $n > 1$,

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 &= \frac{p_{n-2} + a_n (a_{n-1} p_{n-2} + p_{n-3})}{q_{n-2} + a_n (a_{n-1} q_{n-2} + q_{n-3})}
 \end{aligned}$$

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 \end{aligned}$$

Variation of Cassini's Identity

Theorem

If $\frac{p_n}{q_n}$ is the n th convergent to a continued fraction $[a_0, a_1, a_2, \dots]$, then

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

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For $n = 1$, $p_0 = a_0$, $p_1 = a_0 a_1 + 1$, $q_0 = 1$, and $q_1 = a_1$. Thus,

$$p_1 q_0 - p_0 q_1 = (a_0 a_1 + 1) \cdot 1 - a_0 a_1 = 1 = (-1)^0.$$

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$$p_1 q_0 - p_0 q_1 = (a_0 a_1 + 1) \cdot 1 - a_0 a_1 = 1 = (-1)^0.$$

Now, by Induction, for $n > 1$:

$$\begin{aligned} p_n q_{n-1} - p_{n-1} q_n &= (a_n p_{n-1} + p_{n-2}) q_{n-1} - p_{n-1} (a_n q_{n-1} + q_{n-2}) \\ &= p_{n-2} q_{n-1} - p_{n-1} q_{n-2} \\ &= -(-1)^{n-2} \\ &= (-1)^{n-1}. \end{aligned}$$

Example

Consider $\sqrt{6} = [2; \overline{2, 4}]$. Then:

$$\frac{p_0}{q_0} = \frac{2}{1}$$

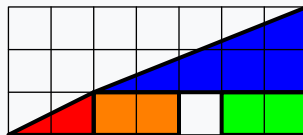
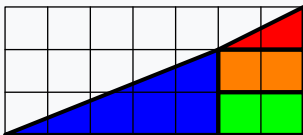
$$\frac{p_1}{q_1} = 2 + \frac{1}{2} = \frac{5}{2}$$

$$\frac{p_2}{q_2} = 2 + \frac{1}{2 + \frac{1}{4}} = 2 + \frac{4}{9} = \frac{22}{9}$$

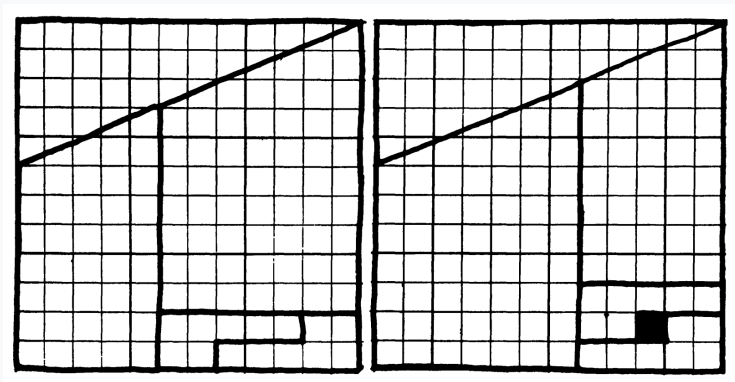
$$\frac{p_3}{q_3} = 2 + \frac{1}{2 + \frac{1}{4 + \frac{1}{2}}} = 2 + \frac{1}{2 + \frac{2}{9}} = 2 + \frac{9}{20} = \frac{49}{20}$$

Example

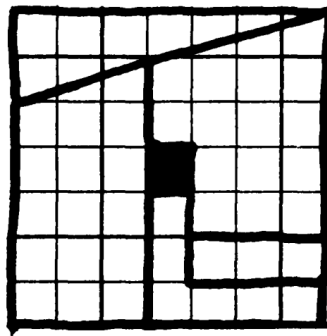
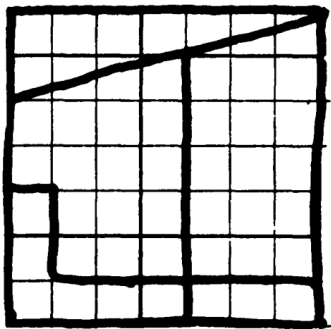
Note that for $\frac{2}{1}$ and $\frac{5}{2}$:



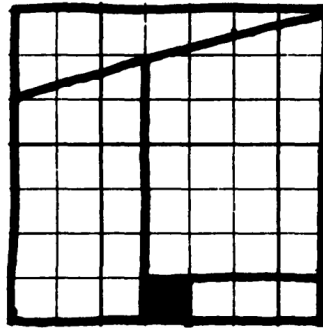
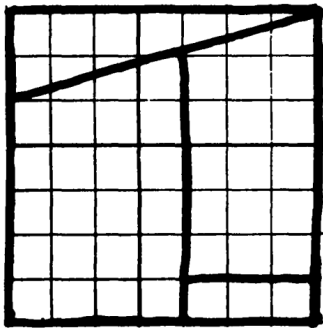
Rectangular



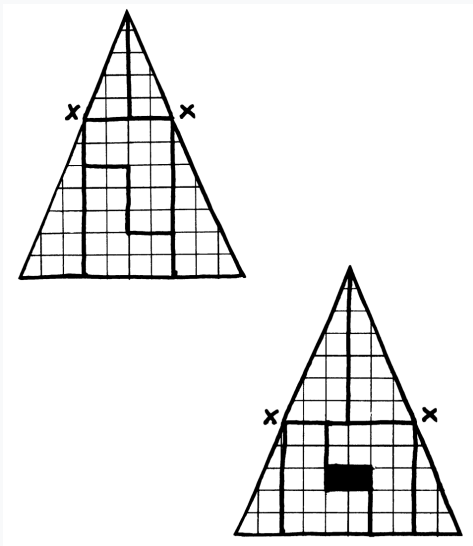
Rectangular



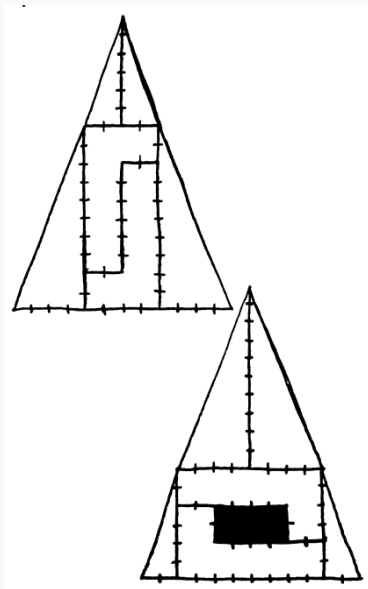
Rectangular



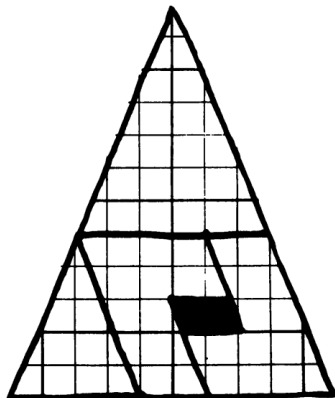
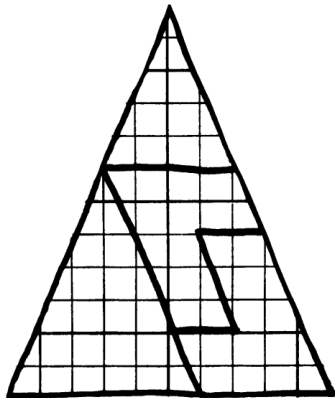
Triangular



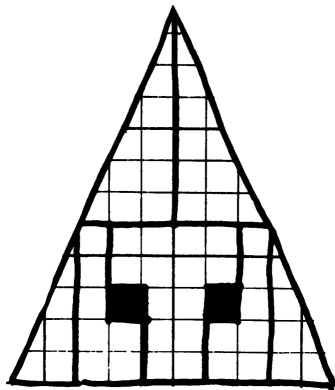
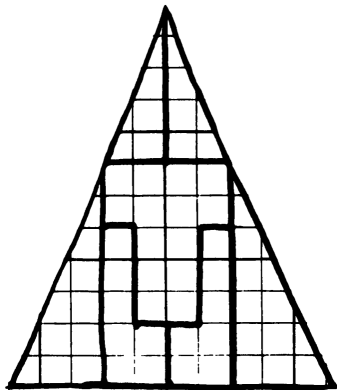
Triangular



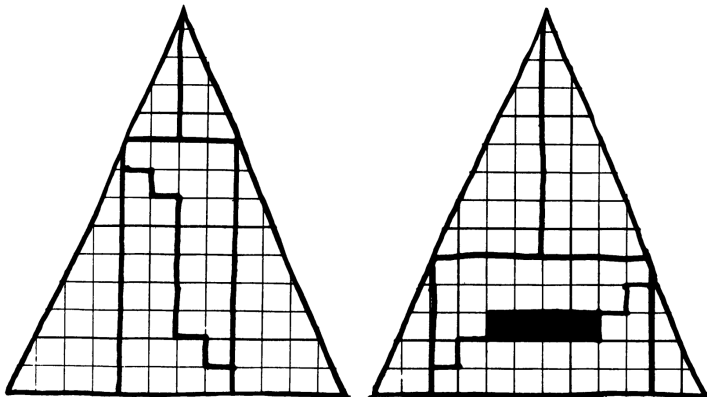
Triangular



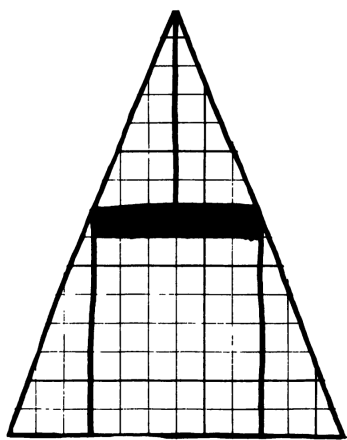
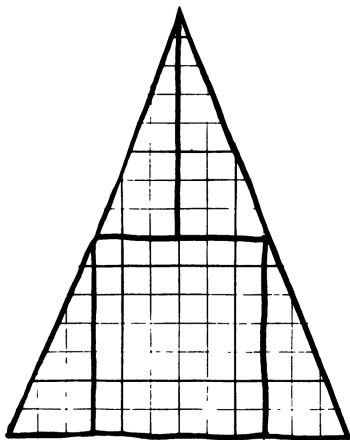
Triangular



Triangular



Triangular



Problems

Problem

Find integers x and y with $1 \leq x, y \leq 50$ such that

$$\sqrt{24} < \frac{x}{y} < 5.$$

Problems

Problem

A triangular number is a positive integer that can be expressed in the form $t_n = 1 + 2 + 3 + \cdots + n$, for some positive integer n . The three smallest triangular numbers that are also perfect squares are $t_1 = 1 = 1^2$, $t_8 = 36 = 6^2$, and $t_{49} = 1225 = 35^2$. What is the sum of the digits of the fourth smallest triangular number that is also a perfect square?

Recall, $t_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} = m^2$. Thus,

$$4n^2 + 4n = 8m^2 \implies (2n+1)^2 = 4n^2 + 4n + 1 = 2(2m)^2 + 1.$$

We want to solve the Pell Equation $a^2 - 2b^2 = 1$.

Problems

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$$\begin{aligned}
 \sqrt{2} &= 1 + (\sqrt{2} - 1) \\
 &= 1 + \frac{1}{\sqrt{2} + 1} \\
 &= 1 + \frac{1}{2 + (\sqrt{2} - 1)} = 1 + \frac{1}{2 + \frac{1}{\sqrt{2} + 1}} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{\ddots}}}}
 \end{aligned}$$

Problems

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$$\frac{1}{1}, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \frac{99}{70}, \frac{239}{169}, \frac{577}{408}.$$

$$a = 577, b = 408 \implies n = 288, m = 204.$$

Problems

Problem

Triangle ABC has side lengths n , $n + 1$, $n + 2$. Interestingly, the median altitude of this triangle is also an integer. Find the sum of the three smallest integer values of n .

Problems

Problem

Show that

$$\sum_{i=1}^n F_i F_{i+1}.$$

is a perfect square.

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